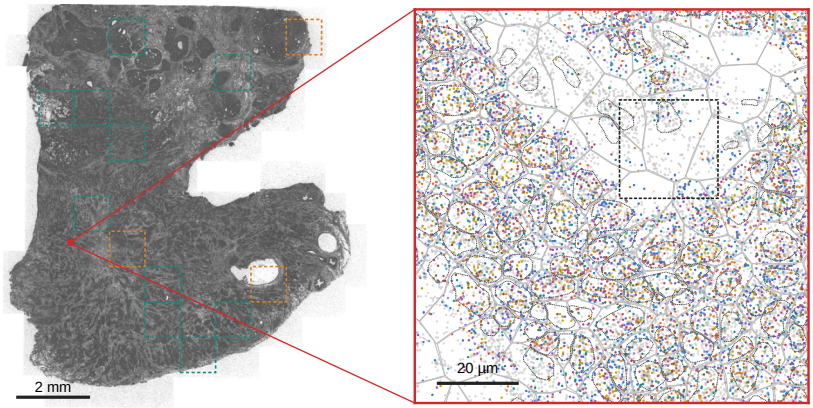
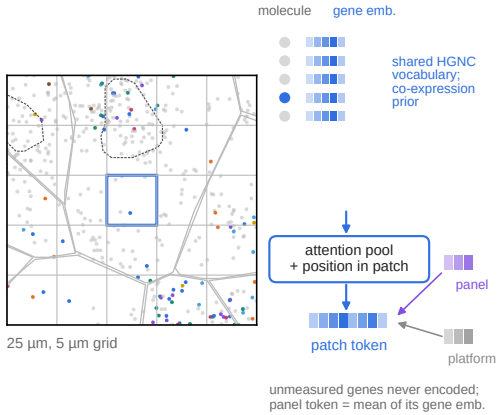
a **Molecules from one Xenium section, no segmentation**



68.1 M molecules · 280 genes · 1 mm blocks:
80% train, 5% validation (orange), 15% probe (teal)

100 µm window; colour = gene (top 8)
outlines: 10x cells / nuclei, eval only

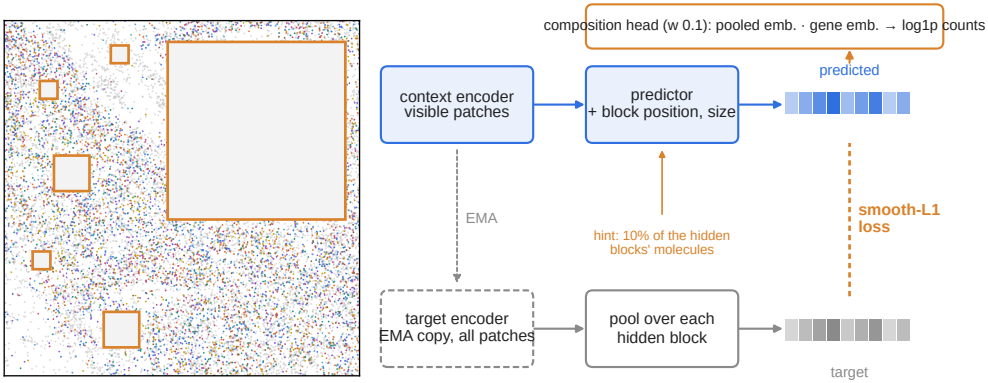
b **5 µm patches become tokens on any panel**



25 µm, 5 µm grid

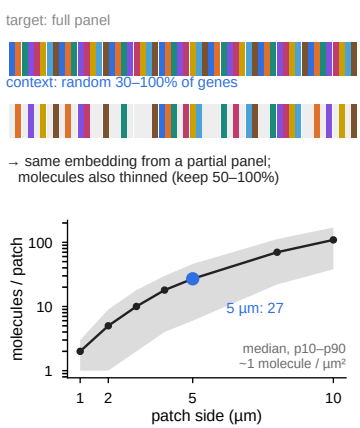
unmeasured genes never encoded;
panel token = mean of its gene emb.

c **Multiscale JEPA: predict hidden regions' embeddings from context**

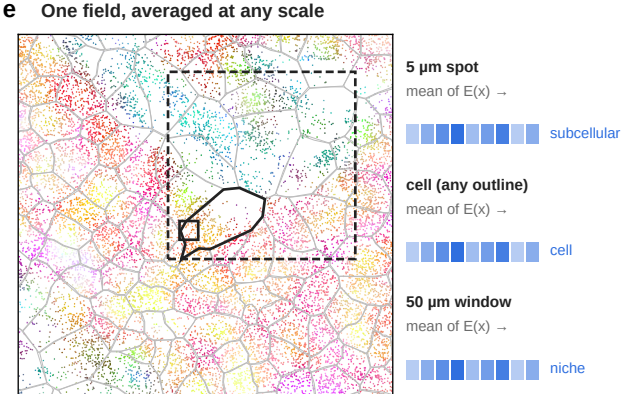


hidden blocks of 5, 10 and 50 µm (training
hides 60–75% of the window; a few shown)

d **Panel dropout and token size**

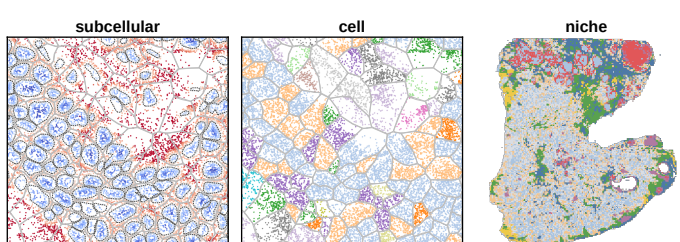


e **One field, averaged at any scale**



schematic of E(x): colours from local composition
(stand-in; the trained field is not shown)

f **Linear probes per scale; the bar is composition at that scale**



distance to nucleus
boundary (blue = inside)

cell type; field averaged
inside each 10x cell

niche labels (shown: 50 µm
composition clusters, k = 8)

Bars to beat: subcellular, comp20 (~2.6 µm composition), measured: nucleus AUROC 0.82 · nucleus-distance ρ 0.62 · cell-distance ρ 0.33; cell, the cell's own counts; niche, CellCompass